

Tech Stack

Frontend (Mobile Apps - Expo Monorepo)

- **Framework:** React Native / Expo SDK 55
- **Styling:** [nativewind](#) (Tailwind CSS for React Native)
- **Motion:** [react-native-reanimated](#)
- **Edge-AI:** [react-native-fast-tflite](#) (for executing the MobileNetV2 models locally)
- **Hardware Comms:** [react-native-ble-plx](#)
- **Offline Data:** [@react-native-async-storage/async-storage](#) & [expo-sqlite](#)
- **Data Visualization:** [react-native-gifted-charts](#)

Backend (API Server)

- **Language & Framework:** Python 3.12+, FastAPI 0.140, Uvicorn
- **ORM:** SQLAlchemy 2.0 with GeoAlchemy2
- **Data Validation:** Pydantic V2
- **Machine Learning:** [xgboost](#) or [tensorflow](#) (for the server-side Smart Pricing model)

Database

- **Engine:** PostgreSQL 16
- **Spatial Extension:** PostGIS 3.4 (Essential for spatial matching logic)

IoT Edge Node (Hardware)

- **Microcontroller:** ESP32
- **Sensors:** Capacitive Soil Moisture Sensor v1.2, DHT22 (Temp/Humidity), Rain Drop Sensor.
- **Language:** C++ (Arduino IDE)

Machine Learning Data & Training Pipeline

- **Base Architecture:** [MobileNetV2](#) (Pre-trained)
- **Target Quantization:** INT8 (for TFLite Edge execution)
- **Primary Training Dataset:** **PlantVillage Dataset** (for foundational leaf feature extraction and baseline disease classification).
- **Real-World Fine-Tuning Dataset:** **PlantDoc Dataset** (for adapting the model to complex, noisy backgrounds and field lighting).
- **Regional Augmentation Dataset:** **Indian Agriculture Crop Disease Dataset** (for specific targeting of local pathogens).
- **Hardware Execution:** Nvidia CUDA/cuDNN optimizations configured for local training execution with batch size restrictions (e.g., [batch_size = 16](#) or [32](#)) to manage VRAM.